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Candidate surname	Other names
USMAN	Grade A* Scholar
Centre Number [9][5][1][0][2]	Candidate Number [0][0][8][8]
Mentora Academy Internal Candidate Ref: MEN-EDX-2026-U4-W09	

Pearson Edexcel International Advanced Level

Chemistry

International Advanced Level

UNIT 4: Rates, Equilibria and Further Organic Chemistry

WEEK 9 ASSESSMENT: STRONG & WEAK ACIDS, pH SCALE, Kw & Ka EQUILIBRIA

Assessment Paper:	WCH14/01 (Week 9 In-Depth Mastery)
Time Allowed:	2 hours 30 minutes
Total Marks:	150 Marks
Topics Examined:	Topic 14A.2: Hydrogen Ion Concentration and the pH Scale (Strong Acids & Bases, Dilutions) Topic 14A.3: Ionic Product of Water, Kw (Thermodynamics, Neutrality at Varying T, Strong Alkalis) Topic 14A.4: Weak Acids, Ka and pKa (Ostwald's Dilution Law, Approximations Breakdown, Quadratic Solutions & Inductive Effects)

INSTRUCTIONS TO CANDIDATES

- Use black ink or ball-point pen. Fill in the boxes at the top of this page.
- Answer ALL questions in Section A, Section B, and Section C.
- In Section A (Questions 1 to 30), select your answer by placing a clear cross in the box [X].
- In Section B and C, show all working in calculation questions. Include appropriate units where required.
- Questions marked with an asterisk (*) are those where your quality of written communication and logical sequence of structure will be assessed.
- A Periodic Table and physical constants are printed on the back page of this booklet.

INFORMATION FOR CANDIDATES

- The total mark for this paper is 150.
- Section A: 30 marks; Section B: 90 marks; Section C: 30 marks.
- Full worked confidential teacher mark scheme is provided at the rear for post-assessment diagnostic.

SECTION A

Answer ALL questions. For each question, make a selection by putting a cross in one box [X]. If you change your mind, put a line through the box and then mark your new choice with a cross.

1 What is the concentration of hydrogen ions, $[H^+]$, in an aqueous hydrochloric acid solution of pH 2.60 at 298 K?

- ☐ **A** $2.51 \times 10^{-3} \text{ mol dm}^{-3}$
- ☐ **B** $3.98 \times 10^{-3} \text{ mol dm}^{-3}$
- ☐ **C** $2.60 \times 10^{-3} \text{ mol dm}^{-3}$
- ☐ **D** $1.58 \times 10^{-2} \text{ mol dm}^{-3}$

(Total for Question 1 = 1 mark)

2 Assuming complete dissociation of both protons, what is the pH of a $0.0250 \text{ mol dm}^{-3}$ aqueous sulfuric acid solution, $H_2SO_4(aq)$, at 298 K?

- ☐ **A** 1.60
- ☐ **B** 1.30
- ☐ **C** 1.00
- ☐ **D** 2.00

(Total for Question 2 = 1 mark)

3 A 10.0 cm^3 sample of $0.100 \text{ mol dm}^{-3}$ hydrochloric acid is diluted with distilled water to a final volume of 1000 cm^3 . What is the pH of the resulting solution at 298 K?

- ☐ **A** 2.00
- ☐ **B** 3.00
- ☐ **C** 4.00
- ☐ **D** 5.00

(Total for Question 3 = 1 mark)

4 What are the units of the ionic product of water, $K_w = [H^+][OH^-]$?

- ☐ A mol dm⁻³
- ☐ B mol² dm⁻⁶
- ☐ C dm⁶ mol⁻²
- ☐ D Dimensionless

(Total for Question 4 = 1 mark)

5 The self-ionisation of water, $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$, has a standard enthalpy change of $H^\circ = +57.1 \text{ kJ mol}^{-1}$. As the temperature of pure water increases, how do K_w and the neutral pH change?

- ☐ A K_w increases; neutral pH increases
- ☐ B K_w decreases; neutral pH decreases
- ☐ C K_w increases; neutral pH decreases
- ☐ D K_w remains constant; neutral pH remains 7.00

(Total for Question 5 = 1 mark)

6 Barium hydroxide, $Ba(OH)_2$, is fully dissociated in aqueous solution. What is the pH of a 0.0150 mol dm⁻³ $Ba(OH)_2$ solution at 298 K? ($K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$)

- ☐ A 12.18
- ☐ B 12.48
- ☐ C 11.82
- ☐ D 13.00

(Total for Question 6 = 1 mark)

7 Which expression represents the acid dissociation constant, K_a , for propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$?

- ☐ A $K_a = [\text{CH}_3\text{CH}_2\text{COO}^-][\text{H}^+] / [\text{CH}_3\text{CH}_2\text{COOH}]$
- ☐ B $K_a = [\text{CH}_3\text{CH}_2\text{COOH}] / ([\text{CH}_3\text{CH}_2\text{COO}^-][\text{H}^+])$
- ☐ C $K_a = [\text{CH}_3\text{CH}_2\text{COO}^-][\text{H}^+]$
- ☐ D $K_a = [\text{CH}_3\text{CH}_2\text{COO}^-][\text{OH}^-] / [\text{CH}_3\text{CH}_2\text{COOH}]$

(Total for Question 7 = 1 mark)

8 A weak monobasic acid X has $K_a = 2.0 \times 10^{-5} \text{ mol dm}^{-3}$, while acid Y has $K_a = 5.0 \times 10^{-4} \text{ mol dm}^{-3}$. Which statement comparing these acids is correct?

- ☐ A Acid X is stronger than acid Y and has a lower pK_a
- ☐ B Acid X is weaker than acid Y and has a higher pK_a
- ☐ C Acid X is stronger than acid Y and has a higher pK_a
- ☐ D Acid X is weaker than acid Y and has a lower pK_a

(Total for Question 8 = 1 mark)

9 What is the pH of a $0.120 \text{ mol dm}^{-3}$ ethanoic acid solution at 298 K, given that $K_a = 1.75 \times 10^{-5} \text{ mol dm}^{-3}$? (Assume standard approximations apply)

- ☐ A 2.84
- ☐ B 2.34
- ☐ C 3.12
- ☐ D 4.76

(Total for Question 9 = 1 mark)

10 When calculating the pH of a weak monobasic acid HA using $[H^+] = \sqrt{K_a \cdot c}$, one assumption is that $[HA]_{eq} \approx c$. Under which condition does this assumption break down most significantly?

- ☐ A When the acid is very concentrated and K_a is very small ($K_a < 10^{-6}$)
- ☐ B When the acid is moderately strong ($K_a > 10^{-2}$) or the solution is very dilute
- ☐ C When the temperature is maintained strictly at 298 K
- ☐ D When the conjugate base forms insoluble precipitates

(Total for Question 10 = 1 mark)

11 A second fundamental assumption made in deriving $[H^+] = \sqrt{K_a \cdot c}$ is that $[H^+]_{eq} \approx [A^-]_{eq}$. What source of hydrogen ions is neglected by this assumption?

- ☐ A Hydrogen ions produced by the auto-ionisation of water
- ☐ B Hydrogen ions absorbed by atmospheric carbon dioxide
- ☐ C Hydrogen ions provided by the undissociated acid HA
- ☐ D Hydrogen ions generated by redox disproportionation

(Total for Question 11 = 1 mark)

12 At 60 °C (333 K), the ionic product of water is $K_w = 9.31 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. What is the pH of pure water at 60 °C, and is the water acidic, neutral, or alkaline?

- ☐ A pH = 6.52; water is slightly acidic
- ☐ B pH = 6.52; water is strictly neutral
- ☐ C pH = 7.00; water is strictly neutral
- ☐ D pH = 7.48; water is slightly alkaline

(Total for Question 12 = 1 mark)

13 Consider the following four carboxylic acids: (I) Ethanoic acid, (II) Chloroethanoic acid, (III) Dichloroethanoic acid, (IV) Trichloroethanoic acid. What is the correct order of INCREASING acid strength (lowest K_a to highest K_a)?

☐ A I < II < III < IV

☐ B IV < III < II < I

☐ C I < III < II < IV

☐ D II < I < III < IV

(Total for Question 13 = 1 mark)

14 2-chlorobutanoic acid is a significantly stronger acid than 4-chlorobutanoic acid. What is the primary physical reason for this difference?

☐ A The inductive electron-withdrawing effect of chlorine decreases with distance from the carboxyl group

☐ B 4-chlorobutanoic acid forms stronger intermolecular hydrogen bonds

☐ C The C-Cl bond in 2-chlorobutanoic acid is ionic rather than covalent

☐ D 2-chlorobutanoic acid has a higher molar mass than 4-chlorobutanoic acid

(Total for Question 14 = 1 mark)

15 A $0.0500 \text{ mol dm}^{-3}$ solution of methanoic acid, HCOOH , has $K_a = 1.80 \times 10^{-4} \text{ mol dm}^{-3}$ at 298 K. What is the degree of dissociation, α , of methanoic acid in this solution?

☐ A 0.0180

☐ B 0.0600

☐ C 0.120

☐ D 0.00300

(Total for Question 15 = 1 mark)

16 If a solution of a weak monobasic acid HA is diluted by a factor of 10 with distilled water, how does the pH change (assuming the approximation $[H^+] = \sqrt{K_a \cdot c}$ remains valid)?

- ☐ A Increases by exactly 1.0 pH unit
- ☐ B Increases by approximately 0.5 pH unit
- ☐ C Decreases by approximately 0.5 pH unit
- ☐ D Remains completely unchanged

(Total for Question 16 = 1 mark)

17 At 298 K, an aqueous solution of sodium hydroxide has a pH of 13.40. What is the concentration of hydroxide ions, $[OH^-]$, in this solution? ($K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$)

- ☐ A $0.251 \text{ mol dm}^{-3}$
- ☐ B $0.400 \text{ mol dm}^{-3}$
- ☐ C $0.100 \text{ mol dm}^{-3}$
- ☐ D $0.0251 \text{ mol dm}^{-3}$

(Total for Question 17 = 1 mark)

18 Which of the following carboxylic acids has the SMALLEST pK_a value at 298 K?

- ☐ A Ethanoic acid (CH_3COOH)
- ☐ B Propanoic acid (CH_3CH_2COOH)
- ☐ C Fluoroethanoic acid (CH_2FCOOH)
- ☐ D Bromoethanoic acid ($CH_2BrCOOH$)

(Total for Question 18 = 1 mark)

19 50.0 cm³ of 0.100 mol dm⁻³ HCl is mixed with 50.0 cm³ of 0.060 mol dm⁻³ NaOH at 298 K. What is the pH of the resulting mixture?

- ☐ A 1.40
☐ B 1.70
☐ C 2.00
☐ D 2.40

(Total for Question 19 = 1 mark)

20 25.0 cm³ of 0.100 mol dm⁻³ barium hydroxide, Ba(OH)₂, is mixed with 25.0 cm³ of 0.150 mol dm⁻³ hydrochloric acid, HCl, at 298 K. What is the pH of the resulting solution? ($K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$)

- ☐ A 12.00
☐ B 12.40
☐ C 12.70
☐ D 1.60

(Total for Question 20 = 1 mark)

21 At 0 °C (273 K), the ionic product of water is $K_w = 1.14 \times 10^{-15} \text{ mol}^2 \text{ dm}^{-6}$. What is the concentration of H⁺(aq) in pure water at 0 °C?

- ☐ A $1.00 \times 10^{-7} \text{ mol dm}^{-3}$
☐ B $3.38 \times 10^{-8} \text{ mol dm}^{-3}$
☐ C $1.14 \times 10^{-8} \text{ mol dm}^{-3}$
☐ D $5.70 \times 10^{-8} \text{ mol dm}^{-3}$

(Total for Question 21 = 1 mark)

22 A weak monobasic acid has $pK_a = 3.75$ at 298 K. What is the numerical value of its acid dissociation constant, K_a ?

- ☐ A $1.78 \times 10^{-4} \text{ mol dm}^{-3}$
- ☐ B $5.62 \times 10^{-4} \text{ mol dm}^{-3}$
- ☐ C $1.00 \times 10^{-3} \text{ mol dm}^{-3}$
- ☐ D $3.75 \times 10^{-4} \text{ mol dm}^{-3}$

(Total for Question 22 = 1 mark)

23 Fluoroethanoic acid ($pK_a = 2.59$) is a stronger acid than chloroethanoic acid ($pK_a = 2.87$). What is the main explanation for this fact?

- ☐ A Fluorine is more electronegative than chlorine, withdrawing electron density more strongly and stabilizing the carboxylate anion
- ☐ B The C-F bond is weaker than the C-Cl bond, making it easier to break
- ☐ C Fluorine has more lone pairs of electrons than chlorine
- ☐ D Fluoroethanoic acid has a higher boiling point than chloroethanoic acid

(Total for Question 23 = 1 mark)

24 According to Ostwald's dilution law, what happens to the percentage dissociation of a weak acid when water is added to its aqueous solution at constant temperature?

- ☐ A Percentage dissociation decreases because dilution suppresses ionisation
- ☐ B Percentage dissociation increases because dilution favours the side with more moles of particles
- ☐ C Percentage dissociation remains constant because K_a is constant
- ☐ D Percentage dissociation first decreases and then remains constant

(Total for Question 24 = 1 mark)

25 A sample of acid rain has a hydrogen ion concentration of $[H^+] = 4.20 \times 10^{-5} \text{ mol dm}^{-3}$. What is the pH of this rainwater?

- ☐ A 4.38
☐ B 4.20
☐ C 5.38
☐ D 3.38

(Total for Question 25 = 1 mark)

26 A student dissolves 0.200 g of sodium hydroxide ($M_r = 40.0$) in distilled water to make 250 cm^3 of solution. What is the pH of this solution at 298 K? ($K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$)

- ☐ A 12.00
☐ B 12.30
☐ C 12.70
☐ D 13.00

(Total for Question 26 = 1 mark)

27 A weak acid HA has $K_a = 2.50 \times 10^{-5} \text{ mol dm}^{-3}$. At what concentration c will the acid be exactly 5.0% dissociated ($\alpha = 0.050$)?

- ☐ A $9.50 \times 10^{-3} \text{ mol dm}^{-3}$
☐ B $5.00 \times 10^{-3} \text{ mol dm}^{-3}$
☐ C $1.00 \times 10^{-2} \text{ mol dm}^{-3}$
☐ D $2.50 \times 10^{-4} \text{ mol dm}^{-3}$

(Total for Question 27 = 1 mark)

28 Which of the following plots will yield a straight line passing through the origin for a weak monobasic acid that obeys the approximation $[H^+] = \sqrt{K_a \cdot c}$?

- ☐ A pH against concentration c
- ☐ B $[H^+]^2$ against concentration c
- ☐ C $\log[H^+]$ against $1/c$
- ☐ D $[H^+]$ against concentration c

(Total for Question 28 = 1 mark)

29 The auto-ionisation of pure liquid ammonia at -50°C is: $2\text{NH}_3(\text{l}) \rightleftharpoons \text{NH}_4^+ + \text{NH}_2^-$. In this solvent system, what is the species analogous to H_3O^+ in water?

- ☐ A NH_2^-
- ☐ B NH_4^+
- ☐ C NH_3
- ☐ D N_2H_4

(Total for Question 29 = 1 mark)

30 A weak acid has $K_a = 4.00 \times 10^{-2} \text{ mol dm}^{-3}$. In a $0.0400 \text{ mol dm}^{-3}$ solution, using the simplified formula $[H^+] = \sqrt{K_a \cdot c}$ gives $[H^+] = 0.0400 \text{ mol dm}^{-3}$ (100% dissociation). Why is this result fundamentally erroneous?

- ☐ A The approximation $[\text{HA}]_{\text{eq}} \approx c$ is invalid because significant dissociation has depleted $[\text{HA}]_{\text{eq}}$
- ☐ B Water auto-ionisation dominates at this concentration
- ☐ C K_a is too small for the simplified formula to apply
- ☐ D The acid behaves as a strong diprotic acid in dilute solution

(Total for Question 30 = 1 mark)

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

31 The pH scale is used as a convenient measure of aqueous hydrogen ion concentration.

(a) Define pH mathematically and explain why Soren Sorensen introduced this logarithmic definition in 1909.

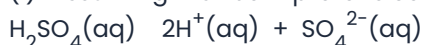
Calculate the concentration of hydrogen ions, $[H^+]$, in a solution of gastric juice with a measured pH of 1.85.

Answer: $[H^+] = \dots\dots\dots \text{mol dm}^{-3}$

(3)

(b) Sulfuric acid, H_2SO_4 , is a diprotic acid. A standard solution is prepared by dissolving 4.904 g of pure anhydrous sulfuric acid ($M_r = 98.08$) in distilled water to make exactly 500.0 cm^3 of solution.

(i) Assuming that both protons completely dissociate in aqueous solution according to:



calculate the concentration of sulfuric acid and the pH of this solution.

Answer: pH =

(3)

(ii) In reality, the first proton is fully dissociated, but the hydrogensulfate ion, HSO_4^- , is a weak acid that only partially dissociates:



Using this equilibrium constant, set up an equilibrium expression to calculate the exact concentration of H^+ and the true pH of this $0.100 \text{ mol dm}^{-3}$ sulfuric acid solution. Calculate the percentage error in $[H^+]$ caused by assuming complete second dissociation.

Answer: True pH = Percentage Error = %

(5)

(c) A titration mixture is prepared by mixing 35.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ nitric acid, HNO_3 , with 65.0 cm^3 of $0.120 \text{ mol dm}^{-3}$ potassium hydroxide, KOH, at 298 K. Calculate the pH of the final mixture. ($K_w = 1.00 \times 10^{-14}$)

mol dm^{-3} at 298 K)

Answer: Final pH =

(4)

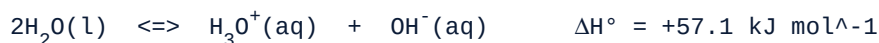
(d) A student dilutes a $0.0500 \text{ mol dm}^{-3}$ solution of hydrochloric acid by adding 90.0 cm^3 of distilled water to 10.0 cm^3 of the acid. Explain, showing all mathematical calculations, the exact effect of this dilution on the pH.

Answer: pH change = units

(3)

(Total for Question 31 = 18 marks)

32 Water undergoes self-ionisation to a very small extent according to the equilibrium:



(a) (i) Write the expression for the equilibrium constant K_c for this reaction, and explain why this expression is simplified to give the ionic product of water, K_w . State the units of K_w .

(3)

(ii) Using the standard enthalpy change $H^\circ = +57.1 \text{ kJ mol}^{-1}$ and Le Chatelier's principle, explain how the value of K_w varies as the temperature of water is increased from 298 K to 373 K.

(2)

(b) At 373 K (100 °C), the value of K_w is $5.13 \times 10^{-13} \text{ mol}^2 \text{ dm}^{-6}$.

(i) Calculate the concentration of hydrogen ions, $[\text{H}^+]$, and the pH of pure boiling water at 373 K.

Answer: $[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$ $\text{pH} = \dots\dots\dots$

(3)

(ii) A student states: 'Because the pH of pure water at 100 °C is less than 7.00, boiling water is acidic and will turn blue litmus paper red.'

Evaluate this statement from a rigorous chemical perspective, explaining why pure water remains strictly neutral at all temperatures.

(3)

(c) Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base that dissociates completely into Ba^{2+} and 2OH^- ions in aqueous solution.

(i) Calculate the pH at 298 K of a solution prepared by dissolving 2.142 g of anhydrous $\text{Ba}(\text{OH})_2$ ($M_r = 171.3$) in distilled water to make exactly 250.0 cm^3 of solution. ($K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 298 K)

Answer: $\text{pH} = \dots\dots\dots$

(4)

(ii) Calculate the volume of distilled water that must be added to 50.0 cm³ of the barium hydroxide solution prepared in (c)(i) to lower its pH to exactly 12.00 at 298 K.

.....

.....

.....

Answer: Volume of water = cm³

(3)

(Total for Question 32 = 18 marks)

33 Ethanoic acid, CH_3COOH , and chloroethanoic acid, CH_2ClCOOH , are weak monobasic acids.

(a) (i) Write the chemical equation for the dissociation of ethanoic acid in aqueous solution and write the expression for its acid dissociation constant, K_a .

(2)

(ii) A $0.150 \text{ mol dm}^{-3}$ solution of ethanoic acid has a measured pH of 2.79 at 298 K. Calculate the values of K_a and pK_a for ethanoic acid at this temperature.

Answer: $K_a = \dots\dots\dots \text{mol dm}^{-3}$ $pK_a = \dots\dots\dots$

(4)

(b) In deriving the simplified expression $[\text{H}^+] = \sqrt{K_a \cdot c}$, two approximations are made:

Approximation 1: $[\text{H}^+]_{\text{eq}} \approx [\text{A}^-]_{\text{eq}}$

Approximation 2: $[\text{HA}]_{\text{eq}} \approx [\text{HA}]_{\text{initial}} = c$

Explain the chemical justification for each approximation and state the conditions under which Approximation 2 becomes invalid.

(4)

(c) Dichloroethanoic acid, CHCl_2COOH , has $K_a = 3.32 \times 10^{-2} \text{ mol dm}^{-3}$ at 298 K. Consider a $0.0500 \text{ mol dm}^{-3}$ solution of this acid.

(i) Calculate the value of $[\text{H}^+]$ and the pH using the simplified formula $[\text{H}^+] = \sqrt{K_a \cdot c}$.

Answer: $[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$ $\text{pH} = \dots\dots\dots$

(2)

(ii) By setting up and solving the exact quadratic equilibrium expression without making Approximation 2:

$$K_a = \frac{[\text{H}^+]^2}{(c - [\text{H}^+])}$$

calculate the true concentration of H^+ and the true pH of this solution.

(4)

(2)

17

34 The degree of dissociation, α , represents the fraction of acid molecules that have ionised in solution.

- (a) (i) For a weak monobasic acid HA with initial concentration $c \text{ mol dm}^{-3}$ and degree of dissociation α , show that the acid dissociation constant can be expressed as:

$$K_a = \frac{c \alpha^2}{(1 - \alpha)}$$

(3)

- (ii) Deduce Ostwald's approximation $\alpha \approx \sqrt{K_a / c}$ when $\alpha \ll 1$. Using this expression, explain why diluting an aqueous weak acid solution increases its percentage ionization.

(3)

- (b) A $0.0200 \text{ mol dm}^{-3}$ solution of methanoic acid, HCOOH , is found to be 8.65% dissociated ($\alpha = 0.0865$) at 298 K.

- (i) Calculate the values of K_a and pK_a for methanoic acid using the exact Ostwald expression.

Answer: $K_a = \dots\dots\dots \text{mol dm}^{-3}$ $pK_a = \dots\dots\dots$

(3)

- (ii) Calculate the degree of dissociation and the pH when this methanoic acid solution is diluted 100-fold to a concentration of $2.00 \times 10^{-4} \text{ mol dm}^{-3}$ at 298 K.

Answer: $\alpha = \dots\dots\dots$ $pH = \dots\dots\dots$

(4)

- (c) The table below lists the pK_a values of five substituted carboxylic acids at 298 K:

Acid	Formula	pK_a at 298 K
Ethanoic acid	CH_3COOH	4.76
Propanoic acid	$\text{CH}_3\text{CH}_2\text{COOH}$	4.87
Fluoroethanoic acid	CH_2FCOOH	2.59
Chloroethanoic acid	CH_2ClCOOH	2.87
Trichloroethanoic acid	CCl_3COOH	0.65

(i) Explain why propanoic acid is a weaker acid than ethanoic acid by referring to the electronic inductive effect of the alkyl group.

(2)

(ii) Explain why fluoroethanoic acid is a stronger acid than chloroethanoic acid, and why trichloroethanoic acid has a much lower pK_a than chloroethanoic acid.

(3)

(Total for Question 34 = 18 marks)

*35 A research student investigated the dissociation constant of an unknown crystalline weak organic acid, X. A mass of 1.832 g of solid acid X was dissolved in distilled water and made up to exactly 250.0 cm³ in a volumetric flask.

- (a) (i) The molar mass of acid X was previously determined by sodium hydroxide titration to be 122.1 g mol⁻¹. Calculate the concentration of the acid solution in mol dm⁻³.

Answer: Concentration = mol dm⁻³

(2)

- (ii) The pH of this solution was measured at 298 K using a calibrated digital pH probe and found to be 2.70. Calculate the values of K_a and pK_a for acid X at 298 K.

Answer: K_a = mol dm⁻³ pK_a =

(3)

- *(b) The student was asked to verify K_a by performing a serial dilution of acid X and measuring the pH of each solution at 298 K.

Describe a full laboratory procedure to:

- Calibrate the digital pH probe using two standard buffer solutions.
- Prepare four serial dilutions (0.0600, 0.0300, 0.0150, and 0.00750 mol dm⁻³) from the original solution using a 25.0 cm³ pipette and a 50.0 cm³ volumetric flask.
- Measure the pH of each diluted solution systematically.
- Identify two significant experimental errors in measuring pH of weak acids and describe how to minimize them.

(6)

- (c) The student plotted a graph of [H⁺]² on the y-axis against the acid concentration [HA] on the x-axis, as shown below:



(i) Explain why the gradient of this straight line is equal to the acid dissociation constant, K_a .

(2)

(ii) Given that the gradient of the student's line of best fit is $6.50 \times 10^{-5} \text{ mol dm}^{-3}$, calculate the pH of a $0.0400 \text{ mol dm}^{-3}$ solution of this acid.

Answer: pH =

(2)

(d) Explain why measuring the pH of an extremely dilute weak acid solution (e.g. $1.00 \times 10^{-6} \text{ mol dm}^{-3}$) left open to the laboratory air leads to a measured pH value noticeably lower than predicted by theory.

(3)

(Total for Question 35 = 18 marks)

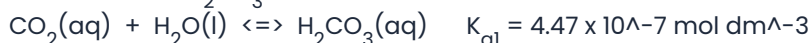
SECTION C

Answer ALL questions. Write your answers in the spaces provided.

36 Atmospheric Chemistry and Ocean Acidification:

Unpolluted natural rainwater is slightly acidic due to the dissolution of atmospheric carbon dioxide. Industrial emissions of sulfur dioxide, SO_2 , form sulfurous acid, H_2SO_3 , leading to acid rain. Dissolved CO_2 also causes ocean acidification.

(a) Atmospheric carbon dioxide dissolves in cloud water according to Henry's law constant $K_{\text{H}} = 3.40 \times 10^{-2} \text{ mol dm}^{-3} \text{ bar}^{-1}$ at 298 K. The partial pressure of CO_2 in air is $4.15 \times 10^{-4} \text{ bar}$. Dissolved CO_2 hydrates to form carbonic acid, H_2CO_3 :



(i) Calculate the equilibrium concentration of dissolved carbonic acid, $[\text{H}_2\text{CO}_3]$, in unpolluted rainwater at 298 K.

Answer: $[\text{H}_2\text{CO}_3] = \dots\dots\dots \text{mol dm}^{-3}$

(2)

(ii) Using $K_{\text{a1}} = 4.47 \times 10^{-7} \text{ mol dm}^{-3}$, calculate the concentration of hydrogen ions and the natural pH of unpolluted rainwater.

Answer: $[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$ $\text{pH} = \dots\dots\dots$

(3)

(b) In an industrial region, rainwater contains sulfurous acid, H_2SO_3 , at a concentration of $2.40 \times 10^{-3} \text{ mol dm}^{-3}$. Sulfurous acid is a moderately strong weak acid with $K_{\text{a1}} = 1.40 \times 10^{-2} \text{ mol dm}^{-3}$ at 298 K.

(i) Write the chemical equation for the first dissociation of sulfurous acid in water.

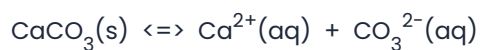
(1)

(ii) Explain why the approximation $[\text{H}_2\text{SO}_3]_{\text{eq}} \approx 2.40 \times 10^{-3} \text{ mol dm}^{-3}$ CANNOT be used in this calculation. By setting up and solving the exact quadratic expression, calculate the true pH of this acid rain.

Answer: Exact pH = $\dots\dots\dots$

(5)

(c) Marine coral reefs are constructed from solid calcium carbonate, $\text{CaCO}_3(\text{s})$, which exists in equilibrium with seawater ions:



As atmospheric CO_2 increases, ocean pH has dropped from 8.20 to 8.08.

Using Le Chatelier's principle and conjugate acid-base equilibria involving CO_3^{2-} and HCO_3^- , explain how ocean acidification leads to the dissolution of calcium carbonate skeletons in coral reefs.

(4)

(Total for Question 36 = 15 marks)

37 Pharmaceutical Chemistry and Human Physiology:

Aspirin (acetylsalicylic acid, 2-acetoxybenzoic acid, $C_9H_8O_4$, $M_r = 180.16$) is a weak monobasic acid with $pK_a = 3.50$ at body temperature (310 K). Non-ionised drug molecules (HA) diffuse across biological lipid membranes much faster than charged ions (A^-).

- (a) (i) Write the chemical equation for the dissociation of acetylsalicylic acid (HA) in water and write the expression for its acid dissociation constant, K_a .

(2)

- (ii) Calculate the value of K_a for aspirin at 310 K.

Answer: $K_a = \dots\dots\dots \text{mol dm}^{-3}$

(2)

- (b) Gastric juice in the human stomach has a resting pH of 1.50, whereas the blood plasma in mucosal capillaries has a physiological pH of 7.40.

- (i) Using the Henderson-Hasselbalch equation:

$$pH = pK_a + \log\left(\frac{[A^-]}{[HA]}\right)$$

calculate the ratio of ionised to non-ionised aspirin, $[A^-] / [HA]$, in the stomach fluid at pH 1.50.

Answer: $[A^-] / [HA]$ in stomach = $\dots\dots\dots$

(3)

- (ii) Calculate the percentage of aspirin that exists in the non-ionised (HA) form in the stomach, and explain why aspirin is rapidly absorbed directly through the stomach lining into the bloodstream.

Answer: Percentage non-ionised = $\dots\dots\dots\%$

(3)

- (c) During intense anaerobic exercise, glycogen is metabolised to lactic acid (2-hydroxypropanoic acid, $CH_3CH(OH)COOH$, $pK_a = 3.86$).

- (i) A muscle biopsy reveals a localized lactic acid concentration of $8.00 \times 10^{-3} \text{ mol dm}^{-3}$. Calculate the pH of an unbuffered aqueous solution of lactic acid at this concentration at 298 K.

Answer: pH = $\dots\dots\dots$

(3)

(ii) In a healthy human athlete, the pH of the blood drops only slightly from 7.40 to approximately 7.25 during vigorous sprinting. Explain briefly why the blood pH does not fall to the value calculated in (c)(i).

(2)

(Total for Question 37 = 15 marks)

DATA SHEET: THE PERIODIC TABLE OF ELEMENTS & PHYSICAL CONSTANTSValues of fundamental physical constants and relative atomic masses (A_r) for Pearson Edexcel Unit 4 Chemistry

Physical Constant / Formula	Symbol / Value	SI Units
Gas Constant	$R = 8.31$	$\text{J K}^{-1} \text{mol}^{-1}$
Ionic Product of Water (298 K)	$K_w = 1.00 \times 10^{-14}$	$\text{mol}^2 \text{dm}^{-6}$
Avogadro Constant	$L = 6.02 \times 10^{23}$	mol^{-1}
Standard Temperature and Pressure	$T = 298 \text{ K}, P = 100 \text{ kPa (1 bar)}$	K, kPa
pH Definition	$\text{pH} = -\log_{10}[\text{H}^+(\text{aq})]$	dimensionless
Acid Dissociation Constant	$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$	mol dm^{-3}
pKa Definition	$\text{pK}_a = -\log_{10}(K_a)$	dimensionless
Henderson-Hasselbalch Equation	$\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$	dimensionless

Periodic Table of the Elements

Group	1	2	3	4	5	6	7	0 / 8
Period 1	H 1.0							He 4.0
Period 2	Li 6.9	Be 9.0	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0	Ne 20.2
Period 3	Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5	Ar 39.9
Period 4	K 39.1	Ca 40.1	Ga 69.7	Ge 72.6	As 74.9	Se 79.0	Br 79.9	Kr 83.8
Period 5	Rb 85.5	Sr 87.6	In 114.8	Sn 118.7	Sb 121.8	Te 127.6	I 126.9	Xe 131.3
Period 6	Cs 132.9	Ba 137.3	Tl 204.4	Pb 207.2	Bi 209.0	Po [209]	At [210]	Rn [222]

CONFIDENTIAL TEACHER MARK SCHEME & EXAMINER TRAP COMMENTARY

Pearson Edexcel International Advanced Level Chemistry – Unit 4 (WCH14/01) – Week 9 Assessment (150 Marks)

Q	Correct Answer	Key Concept & Examiner Guidance	Mark
1	A	$[H^+] = 10^{-pH} = 10^{-2.60} = 2.51 \times 10^{-3} \text{ mol dm}^{-3}$.	1
2	B	Diprotic acid: $[H^+] = 2 \times 0.0250 = 0.0500 \text{ mol dm}^{-3} \Rightarrow pH = -\log(0.0500) = 1.30$.	1
3	B	Dilution factor = $1000/10 = 100$. New $[H^+] = 0.100/100 = 1.00 \times 10^{-3} \text{ mol dm}^{-3} \Rightarrow pH = 3.00$.	1
4	B	$K_w = [H^+][OH^-]$; units = $(\text{mol dm}^{-3}) \times (\text{mol dm}^{-3}) = \text{mol}^2 \text{ dm}^{-6}$.	1
5	C	Endothermic: higher T shifts right, increasing K_w . Thus $[H^+]$ increases and $pH = -\log[H^+]$ decreases.	1
6	B	$[OH^-] = 2 \times 0.0150 = 0.0300 \text{ mol dm}^{-3}$. $[H^+] = 1.00 \times 10^{-14} / 0.0300 = 3.33 \times 10^{-13} \Rightarrow pH = 12.48$.	1
7	A	$K_a = [\text{products}] / [\text{reactant}] = [CH_3CH_2COO^-][H^+] / [CH_3CH_2COOH]$.	1
8	B	Smaller K_a means less dissociated (weaker acid), which gives a higher $pK_a = -\log K_a$.	1
9	A	$[H^+] = \sqrt{K_a \cdot c} = \sqrt{1.75 \times 10^{-5} \cdot 0.120} = 1.449 \times 10^{-3} \Rightarrow pH = 2.84$.	1
10	B	Approximation breaks down when dissociation exceeds ~5%, i.e. when K_a is large or c is very low.	1
11	A	Neglects the auto-ionisation of water ($K_w = [H^+][OH^-]$ produces $10^{-7} \text{ mol dm}^{-3}$ of H^+).	1
12	B	$[H^+] = \sqrt{K_w} = \sqrt{9.31 \times 10^{-14}} = 3.05 \times 10^{-7} \Rightarrow pH = 6.52$. Neutral because $[H^+] = [OH^-]$.	1
13	A	Increasing chlorine substituents increases inductive electron withdrawal, increasing K_a .	1
14	A	Inductive electron-withdrawing effect operates through sigma bonds and diminishes with distance.	1
15	B	$= \sqrt{K_a / c} = \sqrt{1.80 \times 10^{-4} / 0.0500} = \sqrt{3.60 \times 10^{-3}} = 0.0600$ (6.0%).	1
16	B	$[H^+] = \sqrt{K_a \cdot c}$. Diluting 10x divides $[H^+]$ by $\sqrt{10} \approx 3.16$, so pH increases by $\log_{10}(\sqrt{10}) = 0.5$.	1
17	A	$pOH = 14.00 - 13.40 = 0.60 \Rightarrow [OH^-] = 10^{-0.60} = 0.251 \text{ mol dm}^{-3}$.	1
18	C	Fluorine is the most electronegative halogen, so fluoroethanoic acid has the highest K_a and lowest pK_a .	1
19	B	Moles $H^+ = 0.00500$; moles $OH^- = 0.00300$. Excess $H^+ = 0.00200 \text{ mol}$ in $100 \text{ cm}^3 \Rightarrow [H^+] = 0.0200 \Rightarrow pH = 1.70$.	1
20	B	Moles $OH^- = 2 \times 0.025 \times 0.100 = 0.00500$; moles $H^+ = 0.00375$. Excess $OH^- = 0.00125$ in $50 \text{ cm}^3 \Rightarrow [OH^-] = 0.0250 \Rightarrow pOH = 1.60 \Rightarrow pH = 12.40$.	1
21	B	$[H^+] = \sqrt{K_w} = \sqrt{1.14 \times 10^{-15}} = 3.38 \times 10^{-8} \text{ mol dm}^{-3}$.	1
22	A	$K_a = 10^{-pK_a} = 10^{-3.75} = 1.78 \times 10^{-4} \text{ mol dm}^{-3}$.	1
23	A	Higher electronegativity of F withdraws electron density, dispersing negative charge on carboxylate.	1
24	B	Ostwald dilution law: $= \sqrt{K_a / c}$. As c decreases, increases (Le Chatelier shift).	1
25	A	$pH = -\log(4.20 \times 10^{-5}) = 4.38$.	1
26	B	Moles $NaOH = 0.200/40.0 = 5.00 \times 10^{-3} \text{ mol}$. $[OH^-] = 5.00 \times 10^{-3} / 0.250 = 0.0200 \text{ mol dm}^{-3} \Rightarrow pOH = 1.70 \Rightarrow pH = 12.30$.	1
27	A	$K_a = (c \cdot \alpha^2) / (1 - \alpha) \Rightarrow 2.50 \times 10^{-5} = (c \cdot 0.0025) / 0.95 \Rightarrow c = (2.50 \times 10^{-5} \cdot 0.95) / 0.0025 = 9.50 \times 10^{-3} \text{ mol dm}^{-3}$.	1

28	B	$[H^+] = \sqrt{K_a \cdot c} \Rightarrow [H^+]^2 = K_a \cdot c$. Plotting $[H^+]^2$ against c gives a line with gradient K_a .	1
29	B	NH_4^+ is the protonated solvent cation (solvated proton), analogous to H_3O^+ in water.	1
30	A	When K_a is large, dissociation is high, so $[HA]_{eq}$ is significantly less than initial c .	1

Question 31: The pH Scale, Diprotic Acids & Dilutions (18 Marks)

(a) [3 Marks]

- $pH = -\log_{10}[H^+(aq)]$ [1]
- Sørensen introduced logarithmic scale to handle concentrations that span many orders of magnitude (from 1 to 10^{-14} mol dm^{-3}) using manageable small numbers [1]
- $[H^+] = 10^{-1.85} = 1.41 \times 10^{-2}$ mol dm^{-3} (allow 1.4×10^{-2}) [1].

(b)(i) [3 Marks]

- Moles $H_2SO_4 = 4.904 / 98.08 = 0.05000$ mol [1]
- $[H_2SO_4] = 0.05000 / 0.5000 = 0.1000$ mol dm^{-3} [1]
- Assuming full dissociation: $[H^+] = 2 \times 0.1000 = 0.2000$ mol $dm^{-3} \Rightarrow pH = -\log(0.2000) = 0.70$ [1].

(b)(ii) [5 Marks]

- Step 1 dissociation gives 0.100 mol dm^{-3} H^+ and 0.100 mol dm^{-3} HSO_4^- [1]
- Let $x = [H^+]$ from second dissociation: $[H^+] = 0.100 + x$, $[SO_4^{2-}] = x$, $[HSO_4^-] = 0.100 - x$ [1]
- $K_{a2} = ((0.100 + x) \cdot x) / (0.100 - x) = 1.20 \times 10^{-2} \Rightarrow x^2 + 0.112x - 1.20 \times 10^{-3} = 0$ [1]
- Solving quadratic: $x = 9.84 \times 10^{-3}$ mol $dm^{-3} \Rightarrow [H^+]_{total} = 0.100 + 0.00984 = 0.1098$ mol $dm^{-3} \Rightarrow$ True $pH = 0.96$ [1]
- Error = $((0.2000 - 0.1098) / 0.1098) \times 100\% = +82.1\%$ (overestimation of $[H^+]$ by 82%) [1].

(c) [4 Marks]

- Moles $HNO_3 = 0.0350 \times 0.200 = 0.00700$ mol H^+ [1]
- Moles $KOH = 0.0650 \times 0.120 = 0.00780$ mol OH^- [1]
- Excess $OH^- = 0.00780 - 0.00700 = 0.00080$ mol in 100.0 cm^3 (0.100 dm^3) $\Rightarrow [OH^-] = 8.00 \times 10^{-3}$ mol dm^{-3} [1]
- $[H^+] = K_w / [OH^-] = 1.00 \times 10^{-14} / 8.00 \times 10^{-3} = 1.25 \times 10^{-12}$ mol $dm^{-3} \Rightarrow pH = 11.90$ [1].

(d) [3 Marks]

- Initial $[HCl] = 0.0500$ mol $dm^{-3} \Rightarrow$ Initial $pH = -\log(0.0500) = 1.30$ [1]
- Total volume = 10.0 + 90.0 = 100.0 cm^3 (10-fold dilution) $\Rightarrow$ New $[HCl] = 0.00500$ mol $dm^{-3} \Rightarrow$ Final $pH = 2.30$ [1]
- pH increases by exactly 1.00 unit because each 10-fold decrease in $[H^+]$ adds 1 unit to $-\log_{10}[H^+]$ [1].

Question 32: Ionic Product of Water (K_w) & Strong Bases (18 Marks)

(a)(i) [3 Marks]

- $K_c = [H^+][OH^-] / [H_2O]^2$ (or $[H_3O^+][OH^-] / [H_2O]^2$) [1]
- Water is in immense excess and its degree of ionisation is negligible, so $[H_2O]$ is effectively constant [1]
- Combining constant $[H_2O]$ with K_c gives $K_w = [H^+][OH^-]$; units are $mol^2 dm^{-6}$ [1].

(a)(ii) [2 Marks]

- The forward auto-ionisation reaction is endothermic ($H^\circ > 0$) [1]
- By Le Chatelier's principle, increasing temperature shifts the equilibrium in the forward (endothermic) direction to absorb heat, so K_w increases [1].

(b)(i) [3 Marks]

- In pure water, $[H^+] = [OH^-] \Rightarrow K_w = [H^+]^2$ [1]
- $[H^+] = \sqrt{5.13 \times 10^{-13}} = 7.16 \times 10^{-7}$ mol dm^{-3} [1]
- $pH = -\log_{10}(7.16 \times 10^{-7}) = 6.14$ [1].

(b)(ii) [3 Marks]

- Neutrality is defined strictly by the condition $[H^+] = [OH^-]$, NOT by $pH = 7.00$ [1]
- In boiling pure water, every H^+ generated is accompanied by an OH^- ion, so $[H^+] = [OH^-]$ exactly [1]
- Because $[H^+] = [OH^-]$, the solution cannot turn litmus red or exhibit acidic chemical properties [1].

(c)(i) [4 Marks]

- Moles $Ba(OH)_2 = 2.142 / 171.3 = 0.01250$ mol [1]
- $[Ba(OH)_2] = 0.01250 / 0.250 = 0.0500$ mol dm^{-3} [1]

- $\text{Ba}(\text{OH})_2$ fully dissociates to 2OH^- : $[\text{OH}^-] = 2 \times 0.0500 = 0.100 \text{ mol dm}^{-3}$ [1]
- $[\text{H}^+] = 1.00 \times 10^{-14} / 0.100 = 1.00 \times 10^{-13} \text{ mol dm}^{-3} \Rightarrow \text{pH} = 13.00$ [1].

(c)(ii) [3 Marks]

- At target $\text{pH} = 12.00$, $\text{pOH} = 2.00 \Rightarrow \text{Target } [\text{OH}^-] = 0.0100 \text{ mol dm}^{-3}$ [1]
- Initial moles of OH^- in $50.0 \text{ cm}^3 = 0.0500 \text{ dm}^3 \times 0.100 \text{ mol dm}^{-3} = 0.00500 \text{ mol}$ [1]
- Final total volume = moles / target conc = $0.00500 / 0.0100 = 0.500 \text{ dm}^3 = 500 \text{ cm}^3 \Rightarrow \text{Volume of water added} = 500 - 50 = 450 \text{ cm}^3$ [1].

Question 33: Weak Acids: K_a , $\text{p}K_a$ & Quadratic Derivations (18 Marks)

(a)(i) [2 Marks]

- $\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$ (allow with $\text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$) [1]
- $K_a = [\text{CH}_3\text{COO}^-][\text{H}^+] / [\text{CH}_3\text{COOH}]$ [1].

(a)(ii) [4 Marks]

- $[\text{H}^+] = 10^{-2.79} = 1.622 \times 10^{-3} \text{ mol dm}^{-3}$ [1]
- Assuming $[\text{CH}_3\text{COO}^-] = [\text{H}^+]$ and $[\text{CH}_3\text{COOH}]_{\text{eq}} = 0.150 - 0.00162 \approx 0.150 \text{ mol dm}^{-3}$ [1]
- $K_a = (1.622 \times 10^{-3})^2 / 0.150 = 1.75 \times 10^{-5} \text{ mol dm}^{-3}$ [1]
- $\text{p}K_a = -\log_{10}(1.75 \times 10^{-5}) = 4.76$ [1].

(b) [4 Marks]

- Approximation 1 assumes all H^+ comes from HA , justified because water auto-ionisation produces negligible $[\text{H}^+]$ ($10^{-7} \text{ mol dm}^{-3}$) compared to typical weak acid $[\text{H}^+]$ [1]
- Approximation 2 assumes dissociation is negligible so $[\text{HA}]_{\text{eq}} \approx \text{initial conc } c$, justified when K_a is small ($< 10^{-4}$) and initial c is high [1]
- Approximation 2 breaks down when the acid is moderately strong ($K_a > 10^{-2}$) or the solution is very dilute ($c < 10^{-3} \text{ mol dm}^{-3}$), where % dissociation exceeds 5% [2].

(c)(i) [2 Marks]

- $[\text{H}^+] = \sqrt{K_a \cdot c} = \sqrt{3.32 \times 10^{-2} \cdot 0.0500} = \sqrt{1.66 \times 10^{-3}} = 0.04074 \text{ mol dm}^{-3}$ [1]
- $\text{pH} = -\log_{10}(0.04074) = 1.39$ [1].

(c)(ii) [4 Marks]

- $K_a = [\text{H}^+]^2 / (c - [\text{H}^+]) \Rightarrow 3.32 \times 10^{-2} = [\text{H}^+]^2 / (0.0500 - [\text{H}^+])$ [1]
- $[\text{H}^+]^2 + 0.0332[\text{H}^+] - 1.66 \times 10^{-3} = 0$ [1]
- Solving quadratic: $[\text{H}^+] = (-0.0332 + \sqrt{(0.0332)^2 - 4(1)(-1.66 \times 10^{-3})}) / 2 = (-0.0332 + 0.0880) / 2 = 0.0274 \text{ mol dm}^{-3}$ [1]
- True $\text{pH} = -\log_{10}(0.0274) = 1.56$ [1].

(c)(iii) [2 Marks]

- Percentage error = $((0.04074 - 0.0274) / 0.0274) \times 100\% = +48.7\%$ [1]
- Dichloroethanoic acid is over 50% dissociated at this concentration ($0.0274/0.0500 = 54.8\%$), making the assumption $[\text{HA}]_{\text{eq}} \approx c$ completely invalid [1].

Question 34: Degree of Dissociation () & Inductive Effects (18 Marks)

(a)(i) [3 Marks]

- At equilibrium: $[\text{H}^+] = c$, $[\text{A}^-] = c$, $[\text{HA}] = c(1 -)$ [1]
- $K_a = [\text{H}^+][\text{A}^-] / [\text{HA}] = (c \cdot c) / (c(1 -))$ [1]
- Simplifying numerator and denominator: $K_a = (c \cdot) / (1 -)$ [1].

(a)(ii) [3 Marks]

- When $c \ll 1$, $(1 -) \approx 1$, so $K_a \approx c \cdot \Rightarrow \approx \sqrt{K_a / c}$ [1]
- As concentration c decreases (dilution), the denominator in $\sqrt{K_a / c}$ decreases, so must increase [1]
- In Le Chatelier terms: dilution lowers all concentrations, so the equilibrium shifts to the side with more moles of particles (dissociation) [1].

(b)(i) [3 Marks]

- $K_a = (0.0200 \cdot (0.0865)^2) / (1 - 0.0865) = (0.0200 \cdot 0.007482) / 0.9135$ [1]
- $K_a = 1.50 \times 10^{-4} / 0.9135 = 1.638 \times 10^{-4} \approx 1.64 \times 10^{-4} \text{ mol dm}^{-3}$ [1]

• $pK_a = -\log_{10}(1.638 \times 10^{-4}) = 3.79$ [1].

(b)(ii) [4 Marks]

• At $c = 2.00 \times 10^{-4} \text{ mol dm}^{-3}$: $1.638 \times 10^{-4} = (2.00 \times 10^{-4} \times \alpha^2) / (1 - \alpha)$ [1]
 • $\alpha^2 / (1 - \alpha) = 0.819 \Rightarrow \alpha^2 + 0.819 - 0.819\alpha = 0$ [1]
 • $= (-0.819 \pm \sqrt{(0.819)^2 + 4(0.819)}) / 2 = (-0.819 + 1.987) / 2 = 0.584$ (58.4% dissociated) [1]
 • $[H^+] = c = 2.00 \times 10^{-4} \times 0.584 = 1.168 \times 10^{-4} \text{ mol dm}^{-3} \Rightarrow pH = -\log_{10}(1.168 \times 10^{-4}) = 3.93$ [1].

(c)(i) [2 Marks]

- Propanoic acid has a larger alkyl group (ethyl vs methyl) [1]
- Alkyl groups are electron-donating (+I inductive effect), which pushes electron density onto the carboxylate group, destabilising the conjugate base and strengthening the O-H bond, making propanoic acid weaker (higher pK_a) [1].

(c)(ii) [3 Marks]

- Fluorine has a higher electronegativity (4.0) than chlorine (3.0), exerting a stronger electron-withdrawing (-I) inductive effect [1]
- This disperses the negative charge on the carboxylate group more effectively, stabilising the conjugate base and lowering pK_a [1]
- Trichloroethanoic acid has three highly electronegative chlorine atoms withdrawing electron density simultaneously, drastically stabilising the carboxylate anion and reducing pK_a to 0.65 [1].

Question 35: Experimental Determination of K_a (* Level of Response) (18 Marks)

(a)(i) [2 Marks]

- Moles acid $X = 1.832 / 122.1 = 0.01500 \text{ mol}$ [1]
- Concentration $= 0.01500 / 0.2500 = 0.0600 \text{ mol dm}^{-3}$ [1].

(a)(ii) [3 Marks]

- $[H^+] = 10^{-2.70} = 1.995 \times 10^{-3} \text{ mol dm}^{-3}$ [1]
- $K_a = [H^+]^2 / c = (1.995 \times 10^{-3})^2 / 0.0600 = 6.63 \times 10^{-5} \text{ mol dm}^{-3}$ [1]
- $pK_a = -\log_{10}(6.63 \times 10^{-5}) = 4.18$ [1].

*(b) [6 Marks — Level of Response]

- Level 3 (5–6 marks): Comprehensive, logically structured practical protocol. Includes two-point calibration of pH meter using fresh standard buffers (pH 4.00 and 7.00), rinsing electrode with deionised water and blotting dry between buffers. Explains serial dilution: pipette 25.0 cm^3 of $0.0600 \text{ mol dm}^{-3}$ into 50.0 cm^3 flask and make up to mark with deionised water (0.0300 M); repeat consecutively to produce 0.0150 M and 0.00750 M . Details temperature control (thermostatted water bath at 298 K) and waiting for steady digital readout before recording. Identifies two critical errors: temperature fluctuations altering K_a and probe reading, and atmospheric CO_2 dissolving to form carbonic acid in dilute solutions, with practical mitigations (water bath, stoppered flasks).
- Level 2 (3–4 marks): Describes calibration and serial dilution accurately but with minor procedural gaps (e.g. omitting probe rinsing or omitting one dilution step). Identifies at least one error with mitigation.
- Level 1 (1–2 marks): Brief outline of pH measurement and basic dilution with incomplete calibration or error analysis.

(c)(i) [2 Marks]

- From $[H^+] = \sqrt{K_a \times [HA]}$, squaring both sides gives $[H^+]^2 = K_a \times [HA]$ [1]
- Comparing with $y = mx$, plotting $y = [H^+]^2$ against $x = [HA]$ gives a straight line through the origin with gradient $m = K_a$ [1].

(c)(ii) [2 Marks]

- $[H^+]^2 = K_a \times [HA] = (6.50 \times 10^{-5}) \times (0.0400) = 2.60 \times 10^{-6} \text{ mol}^2 \text{ dm}^{-6}$ [1]
- $[H^+] = \sqrt{2.60 \times 10^{-6}} = 1.612 \times 10^{-3} \text{ mol dm}^{-3} \Rightarrow pH = 2.79$ [1].

(d) [3 Marks]

- At $1.00 \times 10^{-6} \text{ mol dm}^{-3}$, theoretical weak acid produces only $\sim 10^{-6} \text{ mol dm}^{-3}$ of H^+ [1]
- Atmospheric CO_2 dissolves in the open beaker, forming carbonic acid: $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$ [1]
- Dissolved CO_2 adds extra H^+ ions ($\sim 2.5 \times 10^{-6} \text{ mol dm}^{-3}$), significantly lowering the measured pH [1].

Question 36: Acid Rain & Ocean Carbonate Equilibria (15 Marks)

(a)(i) [2 Marks]

- $[\text{CO}_2(\text{aq})] = K_H \times P_{\text{CO}_2} = (3.40 \times 10^{-2}) \times (4.15 \times 10^{-4}) = 1.411 \times 10^{-5} \text{ mol dm}^{-3}$ [1]

- Since all dissolved CO_2 is in equilibrium with H_2CO_3 , $[\text{H}_2\text{CO}_3] = 1.41 \times 10^{-5} \text{ mol dm}^{-3}$ [1].

(a)(ii) [3 Marks]

- $[\text{H}^+] = \sqrt{K_{\text{a1}} \times [\text{H}_2\text{CO}_3]} = \sqrt{(4.47 \times 10^{-7}) \times (1.41 \times 10^{-5})} = \sqrt{6.307 \times 10^{-12}}$ [1]
- $[\text{H}^+] = 2.511 \times 10^{-6} \text{ mol dm}^{-3}$ [1]
- $\text{pH} = -\log_{10}(2.511 \times 10^{-6}) = 5.60$ [1].

(b)(i) [1 Mark]

- $\text{H}_2\text{SO}_3(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HSO}_3^-(\text{aq})$ [1].

(b)(ii) [5 Marks]

- $K_{\text{a1}} = 1.40 \times 10^{-2}$ is larger than the concentration $c = 2.40 \times 10^{-3}$, so the acid is predominantly dissociated and $[\text{H}_2\text{SO}_3]_{\text{eq}} \ll c$ [1]
- $K_{\text{a1}} = \frac{[\text{H}^+]^2}{(c - [\text{H}^+])} \Rightarrow 0.0140 = \frac{[\text{H}^+]^2}{(0.00240 - [\text{H}^+])}$ [1]
- $[\text{H}^+]^2 + 0.0140[\text{H}^+] - 3.36 \times 10^{-5} = 0$ [1]
- $[\text{H}^+] = \frac{(-0.0140 + \sqrt{(0.0140)^2 + 4(3.36 \times 10^{-5})})}{2} = \frac{(-0.0140 + 0.01817)}{2} = 2.085 \times 10^{-3} \text{ mol dm}^{-3}$ [1]
- Exact $\text{pH} = -\log_{10}(2.085 \times 10^{-3}) = 2.68$ [1].

(c) [4 Marks]

- Increasing atmospheric CO_2 forms excess H^+ ions in seawater [1]
- Added H^+ reacts with carbonate ions: $\text{H}^+ + \text{CO}_3^{2-} \rightleftharpoons \text{HCO}_3^-$ (protonation to hydrogencarbonate) [1]
- This significantly depletes the concentration of free CO_3^{2-} in seawater [1]
- By Le Chatelier's principle, the equilibrium $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$ shifts to the right, dissolving solid calcium carbonate coral structures [1].

Question 37: Aspirin Pharmacokinetics & Lactic Acidosis (15 Marks)

(a)(i) [2 Marks]

- $\text{C}_9\text{H}_8\text{O}_4(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{C}_9\text{H}_7\text{O}_4^-(\text{aq})$ (or $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$) [1]
- $K_{\text{a}} = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$ [1].

(a)(ii) [2 Marks]

- $K_{\text{a}} = 10^{-\text{p}K_{\text{a}}} = 10^{-3.50}$ [1]
- $= 3.16 \times 10^{-4} \text{ mol dm}^{-3}$ [1].

(b)(i) [3 Marks]

- $\text{pH} - \text{p}K_{\text{a}} = \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) \Rightarrow 1.50 - 3.50 = -2.00$ [1]
- $\frac{[\text{A}^-]}{[\text{HA}]} = 10^{-2.00} = 0.0100$ (or $1/100$) [2].

(b)(ii) [3 Marks]

- Fraction non-ionised = $\frac{[\text{HA}]}{([\text{HA}] + [\text{A}^-])} = \frac{1}{(1 + 0.0100)} = \frac{1}{1.01} = 0.9901 \Rightarrow 99.0\%$ [1]
- In the strongly acidic stomach, 99% of aspirin molecules are uncharged (HA) [1]
- Uncharged, non-polar molecules readily dissolve into and diffuse across the hydrophobic phospholipid bilayer of stomach epithelial cells, facilitating rapid absorption [1].

(c)(i) [3 Marks]

- $K_{\text{a}} = 10^{-3.86} = 1.38 \times 10^{-4} \text{ mol dm}^{-3}$ [1]
- $[\text{H}^+] = \sqrt{K_{\text{a}} \times c} = \sqrt{(1.38 \times 10^{-4}) \times (8.00 \times 10^{-3})} = \sqrt{1.104 \times 10^{-6}} = 1.051 \times 10^{-3} \text{ mol dm}^{-3}$ [1]
- $\text{pH} = -\log_{10}(1.051 \times 10^{-3}) = 2.98$ [1].

(c)(ii) [2 Marks]

- Human blood contains a carbonic acid/hydrogencarbonate buffer system ($\text{H}_2\text{CO}_3 / \text{HCO}_3^-$) [1]
- Added H^+ ions from lactic acid react with excess HCO_3^- ions ($\text{H}^+ + \text{HCO}_3^- \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{CO}_2 + \text{H}_2\text{O}$), neutralizing the acid and preventing a catastrophic drop in blood pH [1].